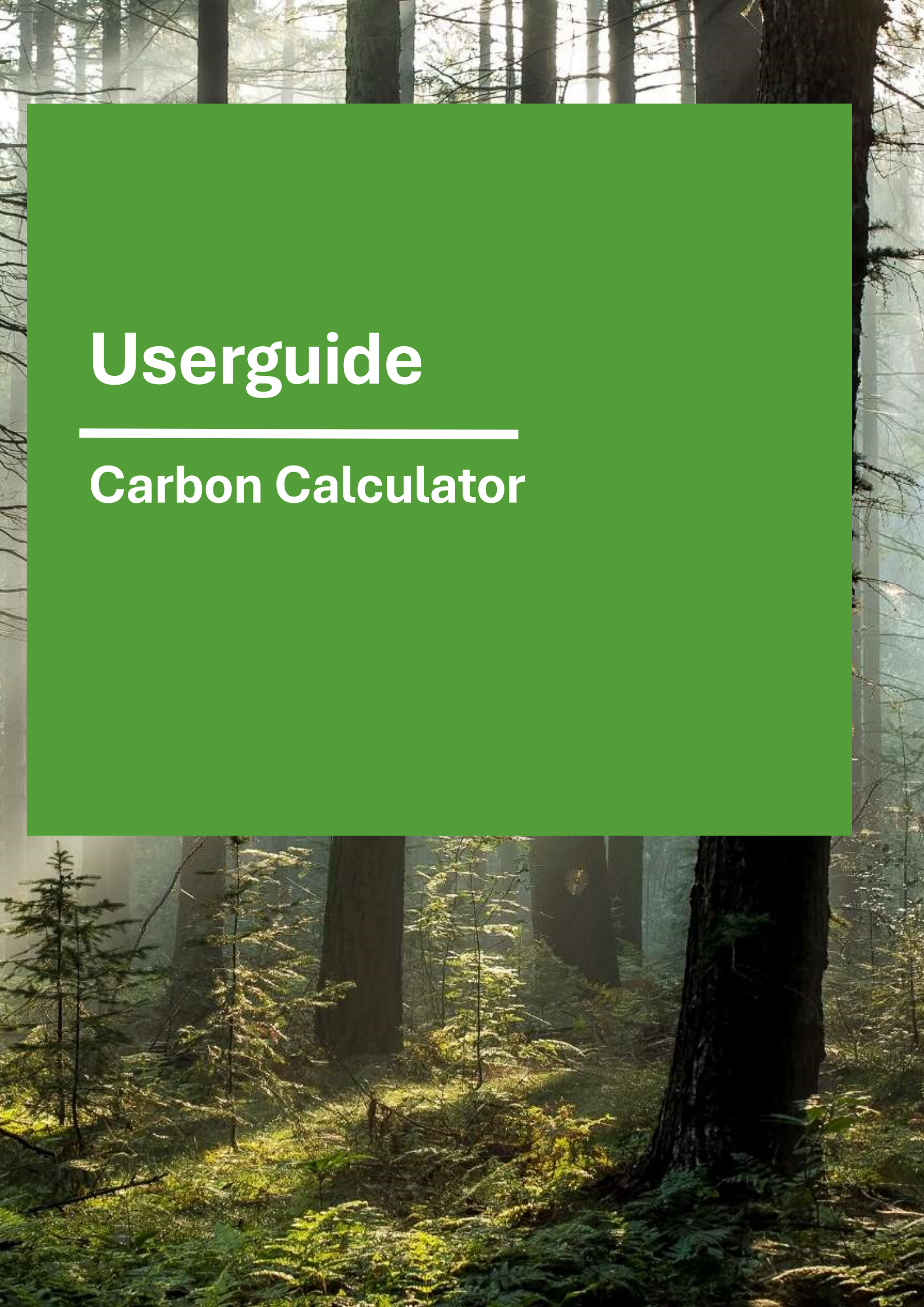


The background of the entire page is a photograph of a forest. Sunlight filters through the trees, creating a dappled light effect on the forest floor. The trees are tall and thin, with some evergreens and some deciduous trees. A large green rectangular overlay covers the upper half of the image, providing a background for the title text.

Userguide

Carbon Calculator

Getting Started

Now that you have received your login credentials, you are ready to use this powerful tool to help your organisation report its greenhouse gas emissions.

This guide will take you through each tab, step-by-step.

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Dashboard



Our Organisation



Report Settings



Add Data



Reports



Reduction Plan

1. Overview

You will see this menu on the left-hand side when you log in to the tool. The rest of the user guide explains each tab here. Follow the steps to use the tool.

Firstly, you will fill out your company details, which will appear in your report in the **Our Organisation** section. Then, we will go to the **Report Settings** tab to enter the parameters of your carbon report.

Then, you will be ready to go into the **Add Data** section and enter your data. Once data is entered, you will visualise it in your **Dashboard** for an overview. For a deeper dive, view the print of the various

reports in the **Reports** section. The next step is to set out the decarbonisation strategy for your business or organisation in the **Reduction Plan** section.

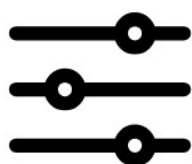
Navigation Summary

1. Our Organisation
2. Report settings
3. Add data
4. Dashboard
5. Reduction Plan



2. Our Organisation

The Our Organisation tab is where you enter your company information; it provides the tool with key contact and location information about the company that will feature in the report.



3. Report Settings

In Report Settings, you will set your reporting boundaries. You will see a bar along the top with four tabs:



Guidelines Tab

The Guidelines tab contains the document you are reading now; you can download it for personal or training use.



Overview Tab

The Overview tab has all the information about the report boundaries. Fill in each box accordingly; each box is explained below.

Reporting Start Date

Choose a reporting start date that aligns with your data. For many companies, this is the financial year. For example, you would choose 01/04/2022, which means you are reporting on data from the period 01/04/2022 – 31/03/2023 for the 2022 reporting year. This will then generate your *Reporting Periods* in the Reporting Periods tab, which we will look at later.

Base Year

The Base Year is a stable reference point for the Reporting Company. Emissions targets will be compared to the Base Year, so this year should remain a stable benchmark. However, base years may be changed or recalculated as the business develops new targets.

Reporting Method

Choose the appropriate reporting method, either location or market-based reporting. *Location-based* reporting reflects the average emissions intensity of grids on which energy consumption occurs, attributing emissions based on geographical location. Alternatively, *market-based* reporting demonstrates a company's choice to procure energy and its associated emissions benefits. It allows companies to claim the benefits

of their specific green power purchases and contracts, such as renewable energy certificates, separate from their physical location's grid emissions.

Organisational Boundaries

These are the limits an organisation sets for accounting and reporting its greenhouse gas emissions. Operational control, financial control, and equity share are the three approaches companies use to determine how greenhouse gas (GHG) emissions from their operations or investments should be included in their carbon inventory.

Net zero

This is the year the company aims to reach net zero, i.e., the emissions produced by the business or organisation are zero or significantly reduced. A net-zero ambition must be aligned with the broader business strategy and stakeholder priorities.

Carbon Reduction Targets

The percentage by which a company or organisation aims to reduce carbon emissions.



Reporting Periods Tab

The reporting period is a specific timeframe for collecting data to track greenhouse gas emissions. If your Reporting Start Date is 01/04/2019, your Reporting Period will run from 01/04/2019 to 31/03/2020 for the Reporting Year 2020. Then, the 2019 Emissions Factors will be used to calculate the emissions and applied to your reported data.

Guidelines Overview **Reporting Periods** Locations

Reporting Year	Reporting Period	Emission Factor Database Year
2020	01/04/2019 - 31/03/2020	2019
2021	01/04/2020 - 31/03/2021	2020
2022	01/04/2021 - 31/03/2022	2021



Locations Tab

If the reporting company or organisation has multiple locations, sites or buildings within its Organisational Boundary, these should be listed here. This calculator will allow you to select locations and compare your emissions across various locations. Click *Create a Location*, then search for your locations and add each one.

This tab will pop up, and fill it in for each site from which you have gathered data. Give each site a location name and description, select whether your company or organisation will include it in the greenhouse gas assessment, and then provide any additional comments that may be relevant to the location.

×

Location Name *

e.g. Head Office

Address *

Description of activities

Describe the activities conducted at this location

GHG assessment inclusion *

Comment

Additional comments or notes



4. Add Data

Go to the Add Data tab. All the emissions categories listed below are divided into scopes 1, 2, and 3. You must decide which ones are relevant to your organisation and its activities.



Company Premises

Company Vehicles



Process Emissions



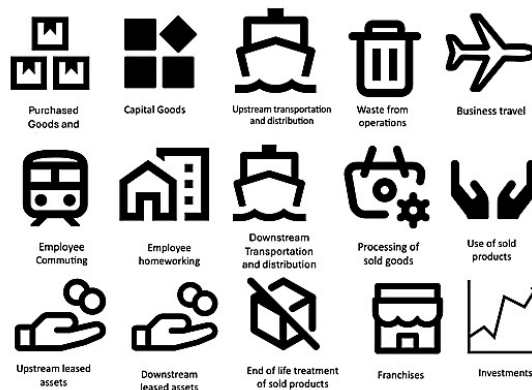
Fugitive Emissions



Purchased Electricity



Purchased Heat & Steam



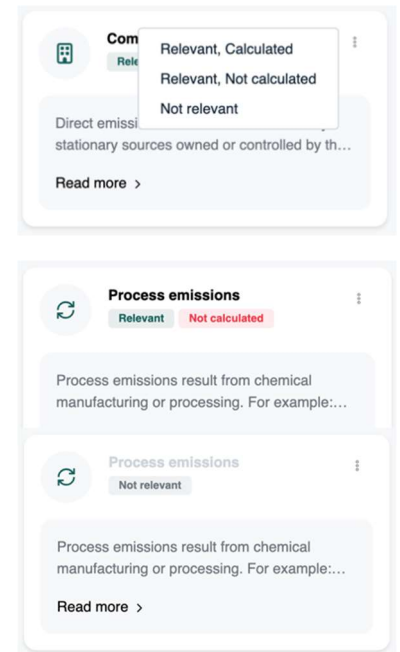
Selecting the relevant emissions categories is a crucial step in the reporting process. It ensures that all relevant emissions are accounted for, providing a comprehensive picture of your organisation's greenhouse gas emissions.

Selecting Emissions Categories

When you have decided on the categories relevant to your organisation, click on the three dots in the top right-hand corner of each emissions category, as shown here.

For the emissions categories you have chosen, select *Relevant, Not Calculated*. To indicate relevant categories for which you haven't yet added the data, the emissions category will now show *Not Calculated* in red, as shown here.

For categories which are **not** relevant to your business or organisation, select 'Not Relevant', and the box will become greyed out and unclickable, as shown on the left here.



Entering your data in each Category

To enter your data, hover over the name of the emissions category and click. You will be taken to a new page. In the top right-hand corner, click Add Data.

Add data

A panel will appear like the one for the Company Vehicles here. Enter the information into each box, starting with a Reference related to your company records. Then, enter the Supplier Name, if relevant. Next, select one of your Locations from those you added earlier.

×

Company Vehicles

Reference

Supplier name

Linked Location

Reporting Year •

Calculation Method •

Type •

Unit of Measurement •

You must select the Reporting Year for this piece of data. Be aware that the year the data is from may differ from the reporting year. For example, an electricity bill from 31/05/2021 will fall within the reporting year 2022. So, be sure to check the dates as shown in the drop-down here.

Reporting Year *

2022 (01/04/2021 - 31/03/2022)
2023 (01/04/2022 - 31/03/2023)
2024 (01/04/2023 - 31/03/2024)
2025 (01/04/2024 - 31/03/2025)

Next, select the Calculation Method, depending on whether you input Liters of fuel, KMs travelled, etc. Then, choose the unit of measurement which should be evident from your data. Then, enter the amount without commas, only with decimal places.

Now select the Activity Data Grade. This is to report the quality of the data you are entering, which is vital to the overall reliability of the report. High-quality activity data is crucial for good GHG reporting, so when entering your data, please follow this grading scale to help you assess its accuracy and reliability. For a guide, see the activity Data Grading Guide in these guidelines.

When all the boxes are filled with the relevant information, click, and you will see a new line of data inside the emissions category.

When you return to the Add Data home page, click the three dots in the right-hand corner and select Relevant, Calculated, to remind you and your colleagues that you have entered data into this Category.

Company Vehicles

Relevant
Calculated

Direct emissions from mobile sources owned or controlled by the organization. For examp...

Read more >

What are custom factors?

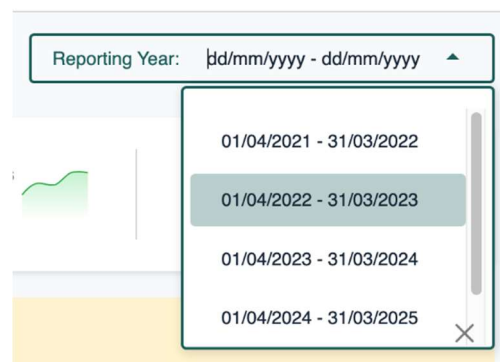
These are emission factors created specifically for a particular process or activity carried out by an organization, rather than using default or standardized emission factors. For example, a company might develop a custom emission factor for a specific manufacturing process that is more efficient and produces fewer emissions than the industry average.



5. Dashboard

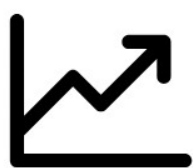
Here, you can view the data that you put into the emissions categories on a dashboard, which gives a complete overview of your carbon emissions. It helps you track, analyse, and manage your environmental impact by consolidating data into easy-to-understand visuals like charts, graphs, and metrics.

You will see Reporting Year and a drop-down menu in the top left. This relates to the information you entered earlier in the Reporting Settings tab. As you select each reporting year from the drop-down menu, the Dashboard will display the data from that year.



Carbon Overview

The Dashboard presents the total carbon emissions measured in tonnes of CO₂ equivalents (CO₂e), offering an overview of your organisation's carbon impact over a specified period. It features a bar graph and a pie chart that visually displays the data for any reporting year. Additionally, there are sections that highlight the Emission Categories contributing the most CO₂e and the specific activities that generate the highest emissions. Lastly, the Dashboard outlines the net zero target and your organisation's carbon reduction goals.



6. Reports

Your data will be presented as reports here; they are all available to download and/or print.



7. Reduction Plan

Here you will find a section on how to make a carbon reduction plan.

8. Data Grading Guide

Activity Data Grades

very good data

Very good data comes from directly measured, reliable primary data from the reporting period or very reliable estimates based on recent operational data. For example, data obtained from monthly electricity meter readings during the reporting year. Volume or no. of sheets of plywood purchased, confirmed by a material accounts record (Purchase orders, delivery notes and process records).

good data

Good data is derived from financial or operational estimations with a justified method. Verified data partly based on assumptions. For example, data based on spend on transportation, with adjustments for weight, distance and broad type of vehicle based on actual journeys.

fair data

Fair data is generally historical data used with justified adjustments to make it reliable or estimates with limited direct measurement. For example, travel data based on two years before the reporting period, extrapolated for 15% growth in employee numbers. Monthly electricity usage based on meter readings from a different but representative site.

basic data

Basic data encompasses broadly allocated estimates with no direct measurement, high uncertainty, and low direct measurement correlation. Spend-based calculations. For example, energy use per employee calculated from UK average for sector, with method/justification uncertain or commuting distance based on averages obtained from Google, with no staff survey or data.

9. Glossary of Terms

Activity: Any process or action that results in the consumption of resources or the generation of emissions. Examples include driving a car, operating machinery, or heating a building. Activities are tracked in carbon footprinting to estimate their associated emissions.

Base Year: A specific year selected as a reference point for measuring future greenhouse gas (GHG) emissions. Organisations choose a base year to create a benchmark for tracking reductions in emissions over time. This helps them set and monitor their progress toward emissions reduction targets consistently and comparably.

Biogenic emissions Are natural gases and particles released from plants, animals, and soil. Key sources include volatile organic compounds (or VOCs) from plants, methane from ruminant animals, and gases from soil bacteria. These emissions affect atmospheric chemistry and are influenced by environmental factors like temperature and sunlight.

Carbon Accounting: A systematic method for tracking, measuring, and reporting carbon emissions within organisations, industries, or regions. It is used to monitor progress toward emissions reduction goals and is often a requirement for regulatory compliance or sustainability reporting.

Carbon Calculating: Measuring or estimating the carbon emissions associated with specific activities, products, or processes. It involves gathering data and applying conversion and emission factors to quantify the greenhouse gases emitted.

Carbon Footprint: The total amount of greenhouse gases (primarily carbon dioxide) emitted directly or indirectly by an individual, organisation, product, or activity, usually expressed in carbon dioxide equivalents (CO₂e). It represents the impact of climate change.

Climate Change: Long-term alterations in global or regional climate patterns, particularly an increase in average global temperatures, primarily due to human activities like deforestation and burning fossil fuels. Climate change leads to various environmental, social, and economic impacts.

Climate Impact Assessment: Encompasses carbon footprinting as part of assessing broader environmental impacts, including carbon emissions.

Conversion Factor: A multiplier used to convert different units of activity data (e.g., litres of fuel consumed, kilowatt-hours of electricity used) into greenhouse gas emissions. It links physical quantities of resources used to their corresponding carbon emissions.

Environmental Footprint: While broader than carbon footprinting, this term sometimes includes carbon as a key component and other ecological impacts, like water and resource use.

Emission: The release of greenhouse gases or other pollutants into the atmosphere. Emissions can come from various human activities, such as burning fossil fuels, industrial processes, or agricultural practices.

Emission Factors: Numerical values used to estimate the emissions produced by an activity, typically measured as emissions per unit of activity (e.g., kg of CO₂ emitted per litre of fuel burned). They help quantify the amount of greenhouse gases released into the atmosphere from specific sources.

Emissions Inventory: A detailed list or database that quantifies the total amount of greenhouse gases emitted by an organisation, sector, or region over a specific period. Emissions inventories are typically used to assess emissions sources and develop strategies for reducing them.

Greenhouse Gas: Gases in the Earth's atmosphere that trap heat and contribute to global warming. The primary greenhouse gases include carbon dioxide (CO₂), methane (CH₄), nitrous oxide (N₂O), and fluorinated gases.

Greenhouse Gas (GHG) Accounting or GHG Footprinting: focuses on calculating and reporting greenhouse gases specifically rather than all types of environmental impacts.

Greenhouse Gas Protocol: A widely used international accounting and reporting standard that guides for organisations to measure, manage, and report their greenhouse gas emissions. It serves as a comprehensive framework for emissions management, including corporate, value chain, and product.

Intensity Ratios: Used in carbon calculation to assess greenhouse gas (GHG) emissions relative to an economic variable like revenue or production output. Unlike absolute emissions, which show total volume, intensity ratios offer context by considering the scale or growth of the organisation and are spend, activity or production-based.